

EUREKA DFM

Design for Manufacturability Analysis

Part ID: Unknown
Date: 2026-06-14 01:08

DFM SCORE

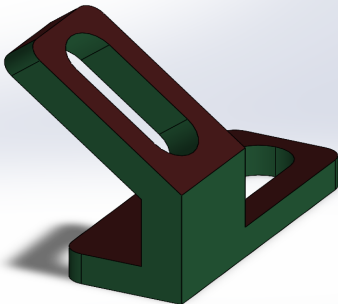
44/100

POOR — SIGNIFICANT REDESIGN
REQUIRED

Manufacturing Process: Injection
Moulding
Selected Material: ABS
GNN Interaction Risk: 0.14
AI Analysis Confidence: 95%

DFM Findings Summary:
● 3 Critical Issues
● 0 Warning Issues
● 1 Info Points

SolidWorks Viewport & Face Highlights



Visual DFM Heatmap & Face Highlights

The CAD viewport screenshot on the left shows the model with real-time face highlights indicating DFM risk areas:

- **RED**: Critical violations (wall thickness or corner fillet radius below limit).
- **ORANGE**: GNN high geometric anomaly risk zones.
- **YELLOW**: Warnings / Informational regions (insufficient draft, GDT control).

Note: These colors are projected live onto the CAD geometry inside the SolidWorks active window.

Critical Findings

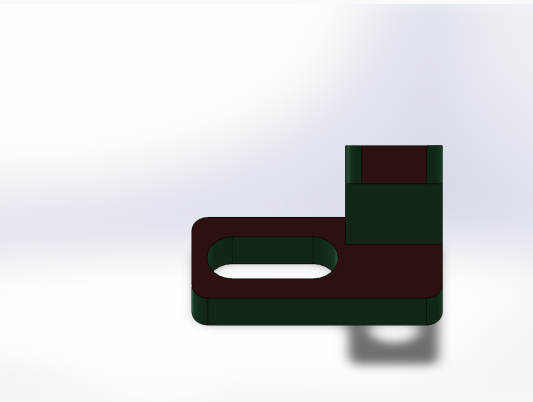
[Severity.CRITICAL] INJ-UNDERCUT-001

Description: Face 2 is undercut -- normal opposes pull direction by 135.0deg. Requires side-action or redesign.

Measured: 135.0deg from pull
Required: <=90deg (no undercut)

Fix: Redesign face to align within 90deg of pull direction, or add a side-action/lifter to the mould tool.

Face 2 is undercut -- normal opposes pull direction by 135.0deg. Requires side-action or redesign.



Face: 2 · Plane · 8.0 mm²

[Severity.CRITICAL] INJ-UNDERCUT-001

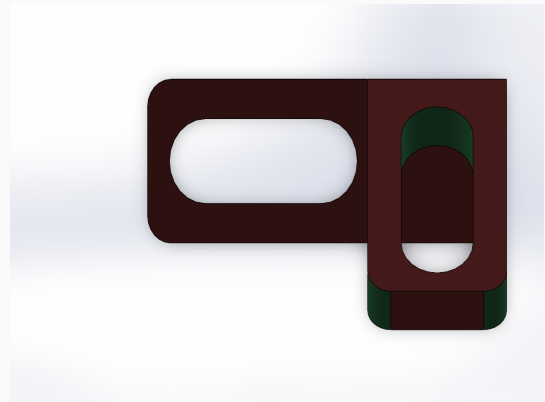
Description: Face 11 is undercut -- normal opposes pull direction by 180.0deg. Requires side-action or redesign.

Measured: 180.0deg from pull

Required: <=90deg (no undercut)

Fix: Redesign face to align within 90deg of pull direction, or add a side-action/lifter to the mould tool.

*Face 11 is undercut -- normal opposes pull direction by 180.0deg.
Requires side-action or redesign.*



Face: 11 · Plane · 57.3 mm²

[Severity.CRITICAL] INJ-UNDERCUT-001

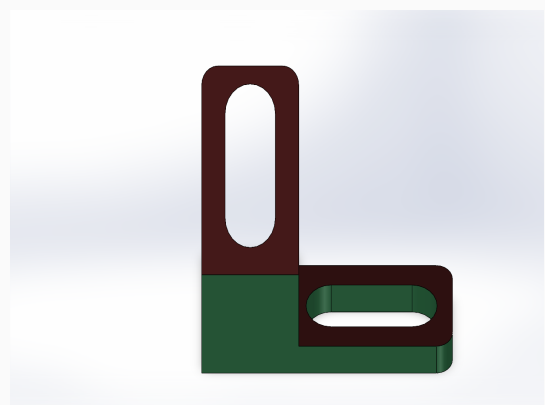
Description: Face 15 is undercut -- normal opposes pull direction by 135.0deg. Requires side-action or redesign.

Measured: 135.0deg from pull

Required: <=90deg (no undercut)

Fix: Redesign face to align within 90deg of pull direction, or add a side-action/lifter to the mould tool.

*Face 15 is undercut -- normal opposes pull direction by 135.0deg.
Requires side-action or redesign.*



Face: 15 · Plane · 40.8 mm²

Informational Findings

[Severity.INFO] GDT-POS-001

Description: Part has 11 holes — true position tolerance required for pattern control and assembly

Measured: 11 holes detected

Required: True position tolerance for hole patterns (≥2 holes)

Fix: Add true position tolerance referencing datums A, B, C

Part has 11 holes — true position tolerance required for pattern control and assembly

Quality Check & Tooling Sign-Off Checklist

- [All critical DFM violations resolved (3 remaining)
]
- [All warning violations reviewed and signed off (0 remaining)
]
- [Material compatibility and flow characteristics confirmed in simulation
]
- [Nominal wall thickness variation is within acceptable tolerance range
]
- [Pull direction and draft angles verified to prevent tool drag
]
- [GNN anomaly interaction risk level below acceptable threshold (0.50)
]